Application of the Satellite-Based Spectral Relationship to the Vertical Localization for the Microwave Humidity Sounder in the Ensemble Kalman Filter

Young-Chan Noh, Eui-Seok Chung, Yonghan Choi, Hyo-Jong Song, Kevin Raeder, and Joo-Hong Kim

***Abstract*— Localization is an essential technique to mitigate the sampling error in the ensemble Kalman filter (EnKF). In order to effectively use the localization function within the EnKF, the specific location information of observations being assimilated is required to measure the distance between the model state variables and the observations. Since the satellite-observed radiance represents integrated quantities across the entire vertical profile of an atmospheric column, it is, however, a challenging issue to accurately assign the vertical location, especially for the satellite radiances sensitive to the variable atmospheric constituents (e.g., water vapor). In this study, we propose an efficient method for assigning the vertical location for observations of microwave humidity sounders (MHSs) onboard low-earth-orbiting (LEO) satellites, using discrete spectral characteristics between the 60 GHz oxygen and 183 GHz water vapor absorption bands. As the radiance differences between the channels are solely used as the predictors for estimating the vertical location in a multivariate regression framework, there is no need to conduct radiative transfer simulations that require additional computation costs. The estimated vertical locations are employed for the vertical localization function within the Data Assimilation (DA) Research Testbed (DART) implementation of ensemble filtering. The verification results show that applying vertical localization to observations from MHSs significantly improves the water vapor analysis derived by the DART system, particularly in the lower troposphere.**

***Index Terms*— Ensemble Kalman filter (EnKF), microwave humidity sounder (MHS), vertical localization.**

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I. Introduction

DATA assimilation (DA) is a crucial component of numerical weather forecasting, used to generate the initial condition "analysis" representing the most accurate state of the Earth's components (e.g., atmosphere, ocean, land, and sea ice) in numerical weather prediction (NWP) system. The initial condition is derived by combining all available observations, including conventional observations (e.g., radiosonde, ship, and aircraft measurements) and satellite observations, with the model forecast "background" using diverse DA techniques [1]. In particular, the accuracy of the initial condition has dramatically improved by assimilating various satellite radiances over regions where conventional ground-based measurements are sparse or absent [2], [3], [4], [5]. Consequently, many operational NWP centers have actively assimilated satellite radiances into their DA systems.

The ensemble Kalman filter (EnKF) differs from the other principal DA algorithm, variational DA, primarily by using the model forecasts from the multiple ensemble members to estimate flow-dependent forecast error covariance between the observations and model state variables [6], [7]. The EnKF, however, suffers from sampling error due to the finite size of the ensemble. In other words, as the ensemble size, which fits in current computer systems, is not enough to capture the true covariance, discrepancies between the estimated covariance and the true covariance could occur, leading to degraded analysis and filter divergence. The sampling error tends to increase as the absolute value of the covariance decreases; thus, to counteract the sampling error in the EnKF, two methods are used to prevent this [8]. "Sampling Error Correction" increases the ensemble spread to account for the uncertainty or imprecision introduced by a small ensemble. "Localization" is applied to reduce the influence of spurious covariance between observations and the model state variables that are far from the observations. A variety of distance-dependent localization functions have been suggested [9], [10], [11]. The "distance" can be more abstract than the euclidean distance, due to factors such as the stratification of the atmosphere, which leads to very different length scales in the horizontal compared to the vertical. The Gaspari–Cohn localization function, which is widely used in the EnKF, is a piecewise continuous fifth-order polynomial approximation to a Gaussian, in which the localization values smoothly decrease from one to zero with the distance between the assimilated observations and the model

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state variables. Such "compactly supported" functions are the most effective at reducing spurious covariance. Then, multiplying the ensemble-derived covariance by these localization values reduces the influence of observations on distant model state variables while preserving their impact on nearby state variables [12], [13].

For the distance-dependent localization functions, it is necessary to correctly define the specific location of the observation, where the maximum localization factor occurs, to measure the physical distance between the observation and the distant model state variable. Satellite radiances are assigned a precise latitude and longitude, like the conventional ground-based observations, but do not have a precise vertical location since radiances represent the vertically integrated quantities over an atmospheric column. The vertical location of a satellite radiance is commonly described by the vertical profile of the weighting function, which is the vertical derivative of the net transmittance between the satellite and each atmospheric layer. An alternative is the sensitivity "Jacobian" of the radiance at the satellite to an atmospheric parameter (e.g., temperature or water vapor) at an atmospheric layer. Both identify the most radiatively active atmosphere layers as seen by the satellite for a particular atmospheric state. This article uses the more familiar weighting functions for illustration and discussion but uses the Jacobians for assigning the vertical locations.

In [14], where the Advanced Microwave Sounding Unit-A and Unit-B (AMSU-A and AMSU-B) radiances were assimilated using the EnKF, the vertical location for the localization function was determined by the pressure level at which the mean weighting function maximum for each channel occurs. As the channels of the AMSU-A instrument are located in the absorption band of oxygen whose concentration is nearly constant, the vertical shapes of their weighting functions are almost unchanged in time and location. Consequently, the predetermined fixed pressure level could be used as the vertical location for the vertical localization function. In contrast, since microwave humidity sounders (MHSs), such as the AMSU-B and the MHS, are sensitive to atmospheric water vapor which varies spatiotemporally, it is unsuitable to use the fixed pressure level derived from the averaged radiance weighting function as the vertical location. Alternatively, for satellite radiances sensitive to variable atmospheric gases (e.g., water vapor and ozone), the radiance sensitivities are obtained through radiative transfer modeling using the spatiotemporally collocated model forecast, including the vertical profiles of these variable gases as the input. Then, the vertical location can be assigned from the simulated radiance sensitivities. An additional computation cost is, however, required to simulate the sensitivities for all satellite radiances sensitive to the variable gases. In the preliminary assimilation experiment, it was found that calculating radiance sensitivities for 10 000 microwave radiance pixels using a radiative transfer model added approximately 24 s/assimilation cycle. Over 200 cycles (equivalent to 50 days), this resulted in a cumulative increase of about 80 min. Additionally, the short-term model forecast error could degrade the accuracy of the simulated sensitivities.

In this study, we attempt to estimate the pressure levels at which the maximum radiance sensitivities (i.e., Jacobian) for the MHS channels occur, based on the spectral relationship between the AMSU-A channels and the MHS channels, without the additional radiative transfer simulation. Since two microwave instruments (i.e., AMSU-A and MHS) are jointly onboard diverse low-earth-orbiting (LEO) satellites such as the MetOp and the National Oceanic and Atmospheric Administration (NOAA) satellites, the radiance data from both instruments can be readily collocated spatiotemporally. As aforementioned, the AMSU-A channels' weighting functions exhibit an almost constant vertical shape because these channels are sensitive to oxygen, the concentration of which is nearly constant. In contrast, the peak height of the radiance sensitivity for the tropospheric MHS channels is likely to vary depending on the amount of and vertical distribution of water vapor in the atmosphere. Considering that within the troposphere, the atmospheric temperature decreases with height, an increase in the peak heights of the radiance sensitivities due to higher water vapor quantity results in a decrease in the MHS radiances, thereby increasing the radiance discrepancy between the lower tropospheric AMSU-A and the tropospheric MHS channels [15]. Consequently, the radiance differences between the lower tropospheric AMSU-A and the tropospheric MHS channels could be used to estimate the peak height of the radiance sensitivity for the MHS channels. In order to assess the assimilation impact of estimated peak heights of the radiance sensitivities for the MHS channels, we apply these estimated peak heights as the vertical locations for the distance-dependent localization function within the EnKF-based DA experiments.

This article is organized as follows. Section II describes the methodology for estimating the peak height of the radiance sensitivity for the MHS channels using a multivariate regression approach. Section III provides background information on the DA system and the satellite observations being assimilated. The setup of the assimilation experiments is presented in Section IV. The results of the impact of the analysis are explored in Section V, followed by a conclusion and discussion in Section VI.

II. Methodology

A. Spectral Relationship

The AMSU-A channels span the spectral absorption band of oxygen, which has a nearly constant concentration in the atmosphere. As a result, the vertical profiles of their radiance weighting functions and sensitivity (e.g., Jacobians) are almost unchanged by the variations in atmospheric variables such as temperature and water vapor. In contrast, the radiative sensitivities of the tropospheric MHS channels tend to shift upward with an increase in water vapor and downward with a decrease in water vapor. This is because the absorption of radiation by water vapor molecules becomes more significant as the concentration of water vapor increases, leading to a strong signal being detected by the tropospheric MHS channels sensitive to mid-to-upper tropospheric water vapor. Considering that the atmospheric temperature within the troposphere tends to decrease with height, an increase (decrease) in the altitude of the radiance sensitivity leads to a decrease (increase) in the measured MHS radiance according to Planck's law, thereby increasing (decreasing) the radiance difference between the

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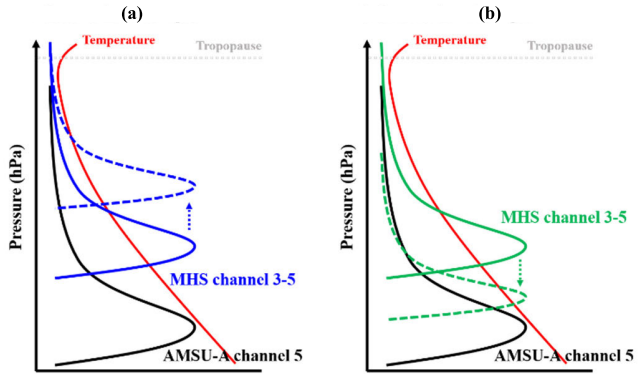


Fig. 1. Vertical variation of radiance weighting function and sensitivity (e.g., Jacobians) for the MHS channels depending on the water vapor quantity in the atmosphere. (a) Increase in water vapor. (b) Decrease in water vapor.

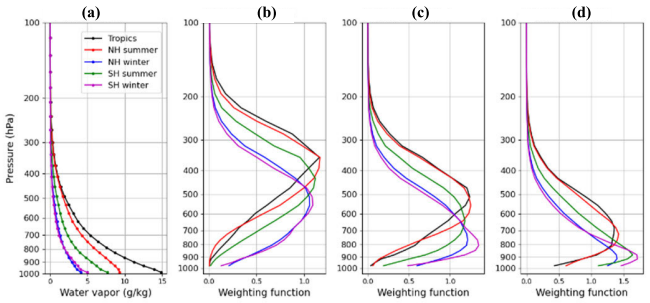


Fig. 2. Vertical profiles of (a) mean specific humidity and (b)–(d) mean weighting functions for the MHS channels 3–5, classified according to the latitude regions and the seasons (tropics, black; Northern Hemisphere during the summer, red; Northern Hemisphere during the winter, blue; Southern Hemisphere during the summer, green; Southern Hemisphere during the winter, magenta).

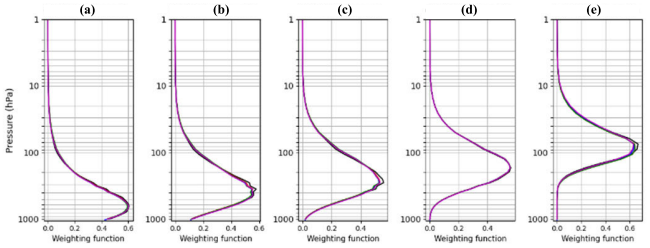


Fig. 3. Mean weighting functions for the AMSU-A. (a) Channel 5, (b) channel 6, (c) channel 7, (d) channel 8, and (e) channel 9, classified according to the latitude regions and the seasons (tropics, black; Northern Hemisphere during the summer, red; Northern Hemisphere during the winter, blue; Southern Hemisphere during the summer, green; Southern Hemisphere during the winter, magenta).

lower tropospheric AMSU-A channels (e.g., channel 5) and the tropospheric MHS channels (Fig. 1). In this study, the peak heights of the radiance sensitivities for the MHS channels are, therefore, estimated using the multivariate regression method which is based on the relationship of the vertical variation of their radiance sensitivities with the radiance differences between the lower tropospheric AMSU-A and the MHS channels.

B. Estimation of the Vertical Location

1) Radiative Transfer Simulation: Spectral outputs (e.g., weighting functions and Jacobians) for the AMSU-A and MHS channels are simulated using the radiative transfer for TIROS Operational Vertical Sounder (TOVS) model version 12 (RTTOV v12), a fast radiative transfer model [16]. The RTTOV simulations are conducted under clear-sky conditions, as only clear-sky AMSU-A and MHS radiances are assimilated in this study. These outputs are used to generate the regression coefficients applied in the multivariate regression method. The AMSU-A instrument has a total of 15 channels that consist of 12 channels (AMSU-A channels 3–14) over the 50–58 GHz oxygen absorption band and three channels (AMSU-A channels 1, 2, and 15) at 23.8, 31.4, and 89 GHz [17], [18]. The MHS instrument also operates in the microwave spectral region, with five frequency channels centered around 89.0, 157.0, 183.311 $\pm$ 1.0, 183.311 $\pm$ 3.0, and 190.31 GHz. These frequencies are chosen for their sensitivity to water vapor in different atmospheric layers [19]. Since MHS channels 1 and 2 are primarily sensitive to the surface, only three tropospheric MHS channels (channels 3–5) are considered for estimating the vertical location. As the input data for the radiative transfer modeling, a set of 20 000 atmospheric profiles from European Center for Medium-Range Weather Forecasts (ECMWFs) forecasts are used, precisely selected to represent global and seasonal atmospheric variability. These datasets are accessible via the NWP Satellite Application Facility (SAF) website (https://www.nwpsaf.eu/site/software/-atmospheric-profile-data/). The reference data provide the temperature and moisture profiles at 137 pressure levels and diverse surface variables (e.g., surface pressure, surface humidity, and skin temperature).

2) Weighting Functions and Jacobians: In order to examine the vertical variation of the weighting functions for the MHS channels 3–5 depending on the water vapor quantity, the averaged weighting functions are calculated for three latitude regions, i.e., Tropics (30°S–30°N), Northern Hemisphere (30°N–90°N), and Southern Hemisphere (30°S–90°S), as well as for two seasons (i.e., June–August and December–February). As shown in Fig. 2(a), the water vapor is abundant in the Tropics. The water vapor quantity in the Northern and Southern Hemispheres shows a marked seasonal variation, with a significantly larger quantity during the summer than during the winter. Three tropospheric MHS channels are designed to provide water vapor information for different atmospheric layers: channel 3 is primarily sensitive to the upper troposphere, channel 4 senses the middle troposphere, and channel 5 focuses on the lower troposphere. The weighting functions peak at different altitudes in the three MHS channels, and their weighting functions tend to shift upward as atmospheric water vapor quantity increases [Fig. 2(b)–(d)]. In other words, the highest peak height of the weighting function for each MHS channel is observed in the Tropics, where water vapor is most abundant. In the Northern and Southern Hemispheres, the peak height of the weighting function is higher during the summer than during the winter. In contrast, the peak heights of the weighting functions for the AMSU-A tropospheric channels 5–9 are almost fixed at the pressure levels to which these channels are sensitive [Fig. 3(a)–(e)].

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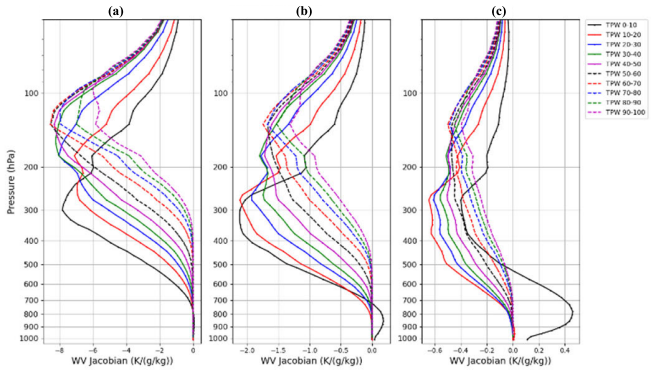


Fig. 4. Mean water vapor Jacobians in units of K/(g/kg) for the MHS. (a) Channel 3, (b) channel 4, and (c) channel 5, for a range of TPW; kg/m^2), over the ocean.

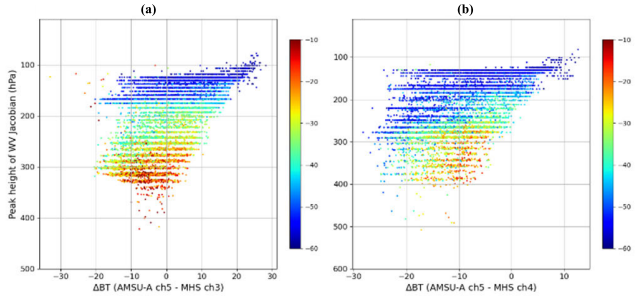


Fig. 5. Scatterplot of the peak heights of water vapor Jacobians versus the difference in the BTs between the AMSU-A channel 5 and the MHS. (a) Channel 3 and (b) channel 4. Color indicates the difference in the BTs between the AMSU-A channels 9 and 5.

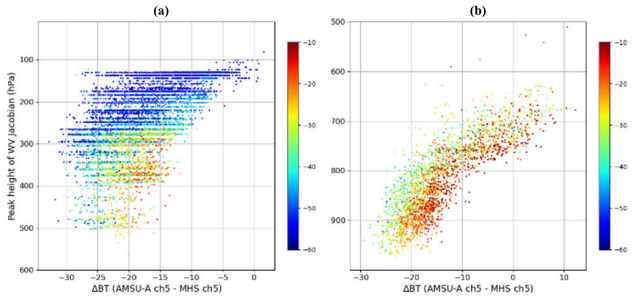


Fig. 6. Scatterplot of the peak heights of water vapor Jacobians versus the difference in the BTs between the AMSU-A channel 5 and the MHS channel 5 for (a) TPW $>$ 10 kg/m^2 and (b) TPW $<$ 10 kg/m^2. Color indicates the difference in the BTs between the AMSU-A channels 9 and 5.

As mentioned in Section II, the peak heights of the radiance sensitivities for the MHS channels are estimated using the multivariate regression method, based on their relationship with the radiance differences between the tropospheric AMSU-A channel and each MHS channel. In this study, the peak height is derived from the water vapor Jacobians, in units of K/(g/kg), of the MHS channels. These indicate how sensitive the radiances are to changes in atmospheric water vapor at a given altitude or pressure level. This choice is based on the fact that the specific humidity (g/kg) is one of the model state variables used to estimate the error covariance with the observations in the EnKF. Only the water vapor Jacobians over the ocean are used since only the MHS observations over the ocean are assimilated in this study.

Fig. 4 illustrates the dependence of the mean water vapor Jacobian on total precipitable water (TPW), kg/m^2), which is estimated by integrating the water vapor quantity in the vertical column of the atmosphere over the ocean. Since the water vapor Jacobian generally has negative values, the peak height is defined as the pressure level at which the water vapor Jacobian exhibits its negative maximum value [Fig. 4(a) and (b)]. Similar to the weighting functions (shown in Fig. 2), the peak heights of water vapor Jacobians for the MHS channels 3 and 4 tend to move upward as the TPW increases. The MHS channel 5, however, exhibits a different pattern of the water vapor Jacobian depending on the TPW. As shown in Fig. 4(c), the maximum value of the water vapor Jacobian has a negative sign when the water vapor is abundant (TPW $>$ 10 kg/m^2), similar to the values for the MHS channels 3 and 4. In a drier atmosphere where the TPW is less than 10 kg/m^2, the water vapor Jacobian maximum is, however, in the lower troposphere and is positive [20]. When the atmosphere dries out, the MHS channel 5 weighting function shifts downward [Fig. 2(d)]. As a result, the lower portion of the water vapor Jacobian is influenced by the surface. Considering that the ocean has a low emissivity at microwave frequencies, the radiance emitted from the lower troposphere is generally larger than that emitted from the ocean surface. Given that the water vapor Jacobian represents the partial derivative, indicating the extent to which a small change in water vapor quantity at a specific altitude affects the radiance measured by the satellite, an increase in water vapor quantity in the lower troposphere, thus, enhances the absorption of radiation by water vapor molecules, thereby reducing the surface-emitted radiance passing through the lower troposphere. The net result of these two effects is a positive value for the water vapor Jacobian [Fig. 4(c)].

3) Multivariate Regression: Figs. 5 and 6 show scatterplots of the peak heights of water vapor Jacobians for the MHS channels against the differences in the brightness temperatures (BTs) between the AMSU-A channel 5 and three MHS channels. The BTs for the AMSU-A channel 5 represent the mid-tropospheric temperature around 600 hPa [Fig. 3(a)]. In contrast, the BTs for the MHS channels decrease as the peak heights of their weighting functions (or water vapor Jacobians) ascend within the troposphere, where the temperature decreases with height (Fig. 1). Consequently, the differences in BTs tend to change from negative to positive with increasing peak heights of the water vapor Jacobians for each MHS channel.

As shown in Fig. 5, the differences in BTs, however, have a significant nonlinear relationship with the peak heights of water vapor Jacobians for the MHS channels 3 and 4. In other words, despite the similarities in BT differences, the corresponding peak heights of water vapor Jacobians are widely distributed between 100 and 400 hPa pressure levels. This nonlinear pattern is also exhibited for the MHS channel 5 when the TPW is larger than 10 kg/m^2 [Fig. 6(a)]; therefore, there are inherent limitations to solely using the BT differences between the AMSU-A channel 5 and each MHS channel in the regression method for estimating the peak heights of

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water vapor Jacobians for the MHS channels. The nonlinear relationship arises because the BT differences between the AMSU-A channel 5 and each MHS channel are affected by both the water vapor quantity and the vertical structure of the temperature in the troposphere; that is, considering that the measured BT is proportional to the mean temperature of the atmospheric layer where the weighting function of each channel covers, the BT differences between the AMSU-A channel 5 and each MHS channel could be different even when the vertical structures of the water vapor are identical if the vertical temperature structures differ. In order to consider the effect of the vertical temperature gradient on the BT difference, the BT difference between the AMSU-A channels 5 and 9 is, therefore, included as an additional predictor in the regression method. The BT difference between AMSU-A channels 5 and 9 provides information on the vertical gradient of tropospheric temperature, as the AMSU-A channels 5 and 9 are sensitive to the lower and upper troposphere, respectively [Fig. 3(a) and (e)].

As shown in Figs. 5 and 6(a), the BT differences between the AMSU-A channels 9 and 5 tend to increase in the negative direction with the peak heights of the water vapor Jacobian. This is because the large negative BT differences between the AMSU-A channels 9 and 5 are mainly distributed in the low latitudes, where the peak heights of water vapor Jacobians are high due to abundant water vapor. When the TPW is less than 10 kg/m^2, the BT differences between the AMSU-A channel 5 and the MHS channel 5, however, have an overall linear relationship with the peak heights of water vapor Jacobians, ranging between 500 and 1000 hPa [Fig. 6(b)].

As the vertical location for the localization function, the peak heights of water vapor Jacobians for the MHS channels are predicted using the multivariate regression method. For the MHS channels 3 and 4, two BT differences are employed as the predictors in the regression method: one is the BT difference between the AMSU-A channel 5 and each MHS channel, and the other is the BT difference between the AMSU-A channels 9 and 5. For the MHS channel 5, similar to the MHS channels 3 and 4, two BT differences are employed in the regression method when the TPW exceeds 10 kg/m^2. When the TPW is below 10 kg/m^2, however, only the BT difference between the AMSU-A channel 5 and MHS channel 5 is used. The predictors regress to the peak heights of water vapor Jacobians for each MHS channel as follows:

$$h_i = \beta_{i,0} + \sum_{j=1}^{N} \beta_{ij} \cdot p_{ij}$$

where $\beta_{i,0}$ indicates the constant component of peak height h_i, and $\beta_{i,j}$ denotes the regression coefficient of the predictor $p_{i,j}$. The subscript i describes the MHS channel number (3–5) and j denotes the predictor (BT differences between the AMSU-A channel 5 and MHS channels, and BT differences between the AMSU-A channel 9 and 5), respectively. In this study, only the MHS observations over the ocean are assimilated within the EnKF-based DA system, so the regression coefficients are calculated using the spectral outputs (e.g., BTs and water vapor Jacobians) simulated from the reference data over the ocean. For the MHS channel 5, the TPW is derived from the BTs of AMSU-A channels 1 and 2, using the retrieval methodology suggested by [21].

TABLE I

ERROR STATISTICS (I.E., MEAN BIAS, ROOT-MEAN-SQUARE ERROR, AND CORRELATION COEFFICIENT) OF THE ESTIMATED PEAK HEIGHTS OF THE WATER VAPOR JACOBIANS AGAINST THE REFERENCE PEAK HEIGHTS OF THESE JACOBIANS FOR THE MHS CHANNELS (3–5)

MHS Ch.	3	4	5	
TPW	-	-	> 10 kg/m^2	10 < kg/m^2
Bias (hPa)	4.17	4.58	1.58	1.90
RMSE (hPa)	24.14	35.44	43.13	46.07
Corr. Coeff.	0.93	0.89	0.82	0.81

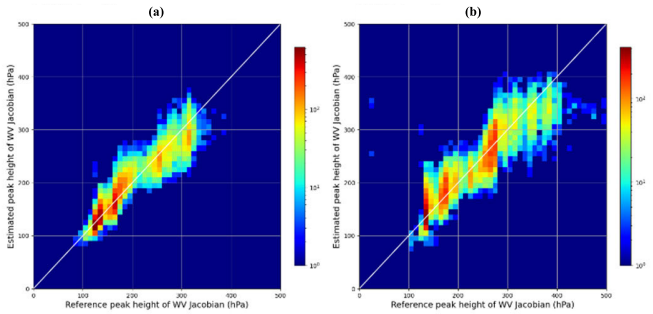


Fig. 7. Scatterplot of estimated peak heights of the water vapor Jacobians versus the reference peak heights of these Jacobians for the MHS (a) channel 3 and (b) channel 4. Colors indicate the number of data points.

Figs. 7 and 8 compare the estimated peak heights of water vapor Jacobians, derived by the regression method, with the peak heights of the reference water vapor Jacobians. For the MHS channels 3 and 4, the estimated peak heights of water vapor Jacobians align well with the reference peak heights (Fig. 7). The biases are less than 5.0 hPa, and the root-mean-square errors (RMSEs) are under 40.0 hPa. The corresponding correlation coefficients are approximately 0.93 and 0.89, respectively (Table I). The accuracy of the regression method is, however, slightly degraded for the MHS channel 5 whose correlation coefficients are approximately 0.80 regardless of the range of TPW (Fig. 8). The RMSEs are about 45 hPa. In particular, it is noted that the estimated peak heights of water vapor Jacobians are higher, ranging between 300 and 400 hPa, when the reference peak heights are distributed from 400 to 500 hPa [Fig. 8(a)]. Considering that the overall correlation coefficients exceed 0.80, it can, nevertheless, be concluded that this multivariate regression method is reliable for estimating the peak heights of water vapor Jacobians for the MHS channels as the vertical location for the localization function within the EnKF-based DA process.

III. DA SYSTEM AND OBSERVATIONS

A. Data Assimilation Research Testbed

Data Assimilation Research Testbed (DART) is a community assimilation software package that was developed by

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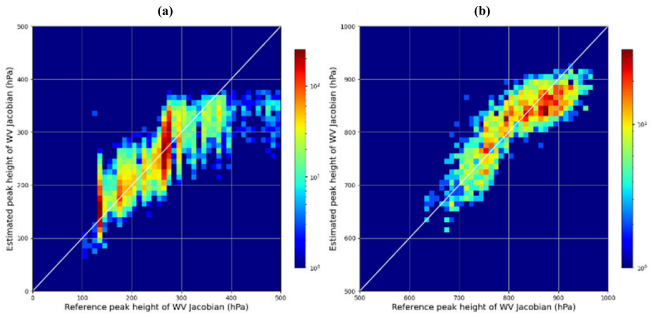


Fig. 8. Scatterplot of estimated peak heights of the water vapor Jacobians versus the reference peak heights of these Jacobians for the MHS channel 5 for (a) TPW $>$ 10 kg/m^2 and (b) TPW $<$ 10 kg/m^2. Colors indicate the number of data points.

the National Center for Atmospheric Research (NCAR) in 2002. DART supports diverse DA techniques such as the EnKF, particle filter, and Quantile Conserving Ensemble Filter Framework. Also, this package includes the processing modules to convert diverse observation data from the native formats into the input format readable in the DART system. DART can be readily adapted to diverse Earth system components (e.g., atmosphere, ocean, land, and sea ice) by the flexible interfaces in the DART package [22], [23]. In this study, the Ensemble Adjustment Kalman Filter (EAKF) is used as the DA scheme. Detailed documentation can be found in [24].

B. Community Earth System Model

Community Earth system model (CESM) version 2 (CESM v2.1.0) is employed as the model component of the EnKF-based DA system. CESM represents a coupled climate–Earth modeling system developed by NCAR, encompassing models for the atmosphere, land surface, ocean, sea ice, land ice, rivers, and waves. These component models can be coupled to exchange states and fluxes [25], [26]. In this study, the atmosphere and land component models are actively coupled, whereas the ocean component (sea surface temperature) and sea ice coverage are predescribed from file-based data. The atmospheric model within CESM2, CAM version 6 (CAM6), is an atmospheric general circulation model (AGCM) featuring the finite volume (FV) dynamical core [27]. CAM6 provides short-term forecasts (6 h) of the atmospheric state, which serve as the background state in the DART assimilation system. The land model used is the community land model version 5 (CLM5). Atmospheric variables are updated directly using information from observations assimilated through the DART system. At the same time, the updated atmospheric state interacts with the land state due to the coupling of the two component models. Both activated models (CAM6 and CLM5) operate at a nominal horizontal resolution of 1° (1.25° in longitude and 0.95° in latitude). CAM6 has 32 vertical levels, extending from the surface to an upper boundary at 3.6 hPa (approximately 40 km). The description of the DART DA system parallels that of [28].

C. AMSU-A and MHS Observations

The AMSU-A and MHS instruments are jointly deployed on the currently operating LEO satellites, which are key components of the Advanced TIROS Operational Sounder (ATOVS), a suite of instruments used for meteorological monitoring from satellites [29]. In this study, the observations of the AMSU-A and MHS instruments onboard three LEO satellites (i.e., Aqua, MetOp-A, and MetOp-B) are employed. The AMSU-A is a cross-track scanning radiometer with a total of 15 channels, whose central frequencies range from around 23.8–89 GHz. The AMSU-A observations provide information on the vertical structure of tropospheric and stratospheric temperature. The instrument scans 30 field-of-views (FOVs) in a swath width of about 2400 km with a spatial footprint size of 48 km in nadir [18]. The MHS instrument is also a cross-track radiometer in which five channels operate at frequencies around 89 GHz and in the range of 157–190 GHz. The instrument has a nominal spatial resolution of 16 km in nadir and provides a total of 90 FOVs per scan line [30]. The scan of the MHS instrument is synchronized with the AMSU-A scan. Prior to the DA, the AMSU-A and MHS observations are collocated spatially and temporally because the TPW, derived from the AMSU-A observations, is used to estimate the peak height of the water vapor Jacobian for the MHS channel 5.

IV. Assimilation Experiment Design

Assimilation experiments are conducted to evaluate how much the assimilation performance is improved by using the estimated peak height of the water vapor Jacobian for the MHS channels as the vertical location for the localization function. Noh et al. [28] demonstrated significant analysis improvement by assimilating the AMSU-A observations within the CESM-DART DA system. In their study, only the horizontal localization function was, however, activated for the AMSU-A observations because the information about the vertical location for the vertical localization was absent; thus, it is also interesting to assess the additional impact on the analysis by activating the vertical localization function at which the mean peak heights of temperature Jacobians for the AMSU-A channels are defined as the vertical locations.

There are four experiments: 1) “AMSUA” where the AMSU-A observations are assimilated but the vertical localization function is not activated; 2) “AMSUA_vert” where the AMSU-A observations are assimilated with the vertical localization function activated; 3) “MHS” where the AMSU-A and MHS observations are assimilated jointly but the vertical localization function is only activated for the AMSU-A observations; and 4) “MHS_vert” where the AMSU-A and MHS observations are assimilated with the vertical localization function activated. The same conventional observations (e.g., ground-based observations and satellite-based winds) are assimilated jointly with the satellite radiance observations for all experiments. Different from satellite observations, horizontal/vertical localization is activated for conventional observations.

Among the 15 channels, only AMSU-A channels 5–11 are assimilated, as the AMSU-A channels 1–4 and 15 are sensitive to the surface, and the upper parts of the weighting functions for the AMSU-A channels 12–14 are above the top of the atmosphere in the CAM6. Since MHS channels 1 and

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TABLE II
DESCRIPTION OF THE EXPERIMENTS USING THE CESM-DART DA. "O" INDICATES PRESENCE/INCLUSION AND "-" PRESENTS ABSENCE/REMOVAL. IN PARENTHESES, THE VERTICAL LOCALIZATION FUNCTION, I.E., THE GASPARI–COHN FUNCTION (GC) OR THE RAMPED BOXCAR FUNCTION (RB), AND THE NORMALIZATION SCALE HEIGHT ARE LISTED

Exp. Name	Conventional	AMSU-A	MHS
AMSUA	O (GC, 1.5)	O (- , -)	-
AMSUA_vert	O (GC, 1.5)	O (RB, 3.0)	-
MHS	O (GC, 1.5)	O (RB, 3.0)	O (- , -)
MHS_vert	O (GC, 1.5)	O (RB, 3.0)	O (RB, 3.0)*

* : The vertical localization function is not applied for the MHS channel 5 observations if the TPW is larger than 10 kg/m^2.

2 are primarily sensitive to the surface, three tropospheric MHS channels (3–5) are assimilated in the CESM-DART DA system. Under clear-sky conditions, AMSU-A and MHS radiances are assimilated over the ocean. Over the land, however, the AMSU-A channels 8–11 are only used. Observation errors are estimated following the method suggested by [31], which accounts for various error sources. These errors include instrument errors, radiative transfer modeling errors, representative errors, and systematic errors. More detailed information on the cloud detection, channel selection, and observation error estimation can be found in [28].

In the AMSUA_vert run, the peak pressures are assigned at 553.9, 353.2, 251.7, 169.0, 74.4, 50.2, and 25.6 hPa, respectively, for the seven AMSU-A channels (channels 5–11), as the vertical locations for the vertical localization function. The preliminary results showed that the water vapor analysis was degraded in the MHS_vert run, where the vertical localization function was applied for the MHS observations, compared with the results from the MHS run. This degradation occurred because the peak heights of water vapor Jacobians for the MHS channel 5 were overestimated when TPW exceeds 10 kg/m^2 [shown in Fig. 8(a)], leading to the misplacement of the vertical localization function; therefore, to mitigate this degradation problem in the MHS_vert run, the vertical localization function is not activated for the MHS channel 5 when the TPW exceeds 10 kg/m^2. A detailed description of four experiments is provided in Table II.

Four experiments are carried out four times a day (0000, 0600, 1200, and 1800 UTC) for the experiment period from 0000 UTC 11 August to 1800 UTC September 30, 2014. The initial ensemble of CAM model states on 0000 UTC August 11, 2014, available at the NCAR Research Data Archive (https://rda.ucar.edu/datasets/ds345.0/), are obtained from the DART reanalysis [23]. 20 ensemble members are integrated to estimate the flow-dependent forecast error covariance between the model state variables and observations being assimilated.

In DART, the distance-dependent localization function is used to remove the spurious covariance induced by the limited ensemble size. The covariance is scaled by a localization factor that decreases from one at the observation location to 0 at some user-selected distance away. DART includes three such functions: the Gaspari–Cohn function, the ramped boxcar function, and the boxcar function [10], [32]. As most conventional observations are measured at single point locations, the Gaspari–Cohn function is widely used, in which the localization factors smoothly decrease from one to zero as the physical distance increases. Considering that the satellite radiance represents the vertically integrated quantity over an atmospheric column, it is, however, more reasonable for the vertical localization factors to maintain a uniform value up to a certain distance and gradually decrease beyond that distance. The localization factor (α) for the ramped boxcar and boxcar functions is 1.0 within the defined distance, and it either gradually decreases or drops to 0 beyond the specified distance as follows:

$$\text{Ramped boxcar } \alpha(d,r) = \begin{cases} 1, & 0 \le d < \dfrac{r}{2} \\ 2\dfrac{r-d}{r}, & \dfrac{r}{2} \le d < r \\ 0, & d \ge r \end{cases}$$

$$\text{Boxcar } \alpha(d,r) = \begin{cases} 1, & 0 \le d < r \\ 0, & d \ge r \end{cases}$$

where d indicates the physical distance between the model state variable and the observations, and r is the localization radius. In order to determine which one is more effective for the satellite radiances between the ramped boxcar and the boxcar, we conducted two additional experiments where the Gaspari–Cohn function was equally applied for the conventional observations, while two different localization functions (i.e., ramped boxcar and boxcar) were used for the satellite observations. A comparison indicates that applying the ramped boxcar resulted in a larger improvement. Consequently, the ramped boxcar function is employed for four experiments.

In DART, the localization half-width can be user-defined, representing half the distance at which the localization factor is zero. As suggested in [28], value of 0.075 rad is used. Since the top height in the model is much lower than the horizontal scale of the Earth, the vertical localization half-width is normalized by a user-defined factor, which defines what vertical scale height is equivalent to 1 rad. The scale height difference between the model top (360 Pa) and the standard surface pressure (101 325 Pa) is 5.64. In this study, a normalization factor of 1.5 is used for the conventional observations. For the AMSU-A and MHS observations, however, a normalization factor of 3.0 is applied to account for the vertically broad weighting functions of the satellite channels.

V. RESULTS

In order to assess the impact of the vertical localization for the satellite radiances on the analyses derived from the experiments, the analyses are validated against the ECMWF reanalysis version 5 (ERA5). The ERA5 reanalysis is generated through the assimilation of all available observation data within the ECMWF operational NWP system and provides consistent maps without spatial gaps, in particular, in the

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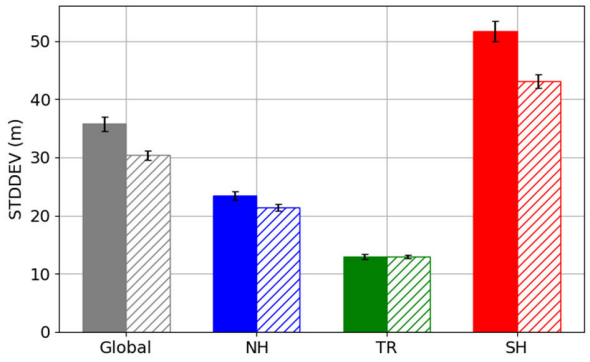


Fig. 9. Standard deviation of the analysis of 500 hPa GPH over the global (gray), Northern Hemisphere (NH, blue); tropics (TR, green); and Southern Hemisphere (SH, red); quantified against the ERA5 reanalysis. Filled and hatched bars indicate the results for the AMSUA and the AMSUA_vert runs, respectively. The 99% confidence intervals are indicated by the vertical black lines.

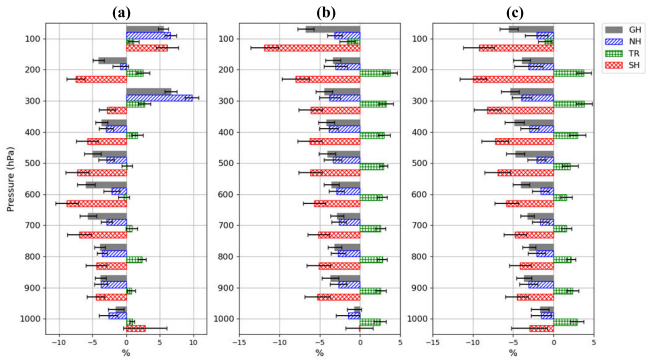


Fig. 10. Normalized difference of the standard deviation of the analysis of (a) temperature, (b) zonal wind, and (c) meridional wind between the AMSUA_vert and the AMSUA runs, quantified against the ERA5 reanalysis. Hatched colors indicate the latitude regions (global, gray; Northern Hemisphere, blue; tropics, green; and Southern Hemisphere, red). Horizontal lines indicate 99% confidence intervals.

Southern Hemisphere, where the conventional observations are sparse. For this reason, the ERA5 is widely used to evaluate model-derived output.

A. Analysis Impact of Vertical Localization for AMSU-A

Fig. 9 shows the standard deviation of 500 hPa geopotential height (GPH) for the AMSUA and the AMSUA_vert runs relative to the ERA5 reanalysis, over four latitudinal domains. They are generally larger in the Southern Hemisphere than in other latitude regions. The limited number of conventional observations assimilated in the Southern Hemisphere, where a significant portion is covered by ocean, results in reduced analysis accuracy, compared to the other two regions (i.e., Tropics and Northern Hemisphere). The global-mean standard deviation decreases to about 30 m for the AMSUA_vert run, compared to the error (about 36 m) for the AMSUA run. In particular, the reduction is more pronounced in the Southern Hemisphere. This suggests that using fixed pressure levels as the vertical locations for the AMSU-A observations and applying vertical localization function leads to improved use of satellite radiances in observation-sparse regions.

In addition to the errors of 500 hPa GPH, we assess the analysis in terms of primary atmospheric variables (i.e., temperature, zonal wind, and meridional wind). Fig. 10 presents the normalized difference in the standard deviation of temperature, zonal wind, and meridional wind between the AMSUA_vert and the AMSUA runs, across four latitude regions (i.e., global, Northern Hemisphere, Southern Hemisphere, and Tropics). Since the difference in standard deviation is normalized by the standard deviation for the AMSUA run, a negative value indicates that the vertical localization function improves the analysis in the AMSUA_vert run in comparison to the AMSUA run. Conversely, a positive value suggests that the analysis error increases in the AMSUA_vert run, indicating a negative impact from the vertical localization.

For temperature, the AMSUA_vert run shows a substantial reduction in the analysis errors in the entire troposphere and lower stratosphere except for the 100 and 300 hPa pressure levels, compared to the AMSUA run. In particular, the global-mean errors decrease significantly near the middle troposphere, with reduction of 6% at 600 hPa. Similar to the results of 500 hPa GPH, a large error reduction mainly occurs in the Southern Hemisphere, where the errors decrease up to about 8% at 600 hPa [Fig. 10(a)]. Since the conventional observations are sparse in the Southern Hemisphere, the assimilation impact of satellite observations becomes more significant compared to the other regions (i.e., Tropics and Northern Hemisphere) where the conventional data are relatively abundant. For two wind components (i.e., zonal wind and meridional wind), similar to the results for the temperature, the global-mean analysis errors for the AMSUA_vert run tend to decrease by 3%–7% throughout the entire troposphere and lower stratosphere [Fig. 10(b) and (c)]. In the Southern Hemisphere, the error is reduced by about 12% for the zonal wind at 100 hPa and by about 10% for the meridional wind at 200 hPa. In the Tropics, the errors, however, slightly increase by 2% for both wind components. Given that the model error covariance accounts for the multivariate correlation between different model state variables, a change in one model variable can influence another variable during the DA process. Since model variables are, moreover, interconnected through the governing equations and physical parameterizations, a change in one model variable leads to adjustments in other variables during the model time integration. The change in the wind components is, therefore, an indirect impact of the enhanced temperature field achieved through the assimilation of AMSU-A observations.

B. Impact of Vertical Localization for MHS on Moisture Analysis

Before assessing the impact of the vertical localization for the MHS observations on the moisture analysis, it would be valuable to evaluate how much assimilating the MHS observations contributes to improving the moisture analysis within the CESM-DART system. In the comparison between the MHS and the AMSUA_vert runs, the moisture analysis is significantly enhanced by assimilating the MHS observations, even without using the vertical localization function for the

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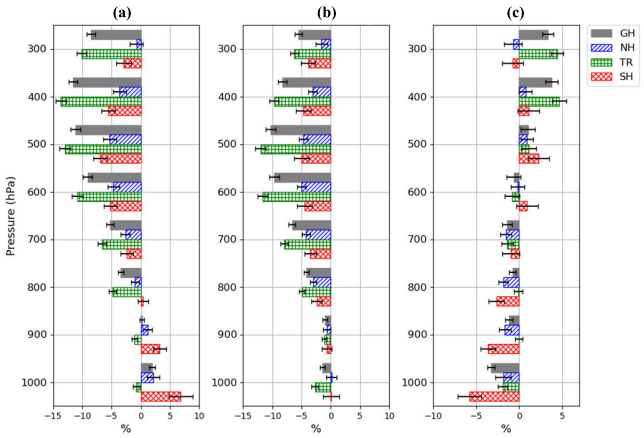


Fig. 11. Normalized difference of the standard deviation of specific humidity analysis, quantified against the ERA5 reanalysis, between (a) MHS and the AMSUA_vert runs, between (b) MHS_vert and the AMSUA_vert runs, and between (c) MHS_vert and MHS runs. Hatched colors indicate the latitude regions (global, gray; Northern Hemisphere, blue; tropics, green; and Southern Hemisphere, red). Horizontal lines indicate 99% confidence intervals.

MHS observations [Fig. 11(a)]. Large error reduction occurs mainly in the middle and upper troposphere, between 300 and 700 hPa, especially in the Tropics, where the error decreases by about 14% at 400 hPa. In the Southern Hemisphere, the error, however, increases by 3%–7% in the lower troposphere (900–1000 hPa), while error reductions are observed in the middle and upper troposphere. The MHS channel 5 is primarily sensitive to the water vapor in the middle and lower troposphere, but it also has a sensitivity to the surface. In order to minimize the negative impacts induced by uncertain surface parameters (i.e., surface temperature and spectral emissivity), we only assimilate the MHS observations over the ocean, where these surface parameters are better constrained compared to the land surface. The assimilation impact is, nevertheless, negative in the lower troposphere. The degraded impact seems to be due to the dual sensitivity of MHS channel 5, which induces spurious correlations between the model variables and the observations.

In order to evaluate the additional analysis impact of using the vertical localization function for MHS observations, the normalized difference in the standard deviation of specific humidity between the MHS_vert and the AMSUA_vert runs is calculated [Fig. 11(b)]. The errors are significantly reduced in the middle and upper troposphere (300–700 hPa), with the most notable improvement occurring in the Tropics, where the error reduction reaches about 12% at 500 hPa. Compared to the results in Fig. 11(a), the positive impact slightly decreases in the upper troposphere (300–400 hPa) when the vertical localization function is activated for the MHS observations. Activating the vertical localization function, however, results in an improvement in the lower troposphere (900–1000 hPa).

These error patterns are clearly exhibited in the comparison between the MHS_vert and the MHS runs [Fig. 11(c)]. The global-mean analysis error for the MHS_vert run increases above 500 hPa, reaching a maximum of about 3% at 400 hPa, while it decreases up to a maximum of about 6% below 500 hPa. In both the Northern and Southern Hemispheres, significant error reductions are exhibited in the lower troposphere. In particular, the error reduction reaches about 6% in the Southern Hemisphere at 1000 hPa. Although a negative impact is observed in the middle and upper troposphere, the positive impact on the lower troposphere is significant, given that the majority of atmospheric water vapor is concentrated in the lower troposphere [shown in Fig. 2(a)]. These results, thus, support that the additional analysis benefit is obtained by applying the vertical localization function at which the vertical location is estimated using the regression method developed in this study.

VI. Conclusion and Discussion

In this study, we attempt to estimate the peak height of the water vapor Jacobian for the MHS observations using the multivariate regression method, in which the radiance differences between the MHS and AMSU-A channels are employed as the only regression predictors. Then, the estimated peak heights are used as the vertical locations for the vertical localization function in the EnKF.

Since the AMSU-A channels are distributed over the absorption band of oxygen, which is a stable component of the atmosphere, their radiance weighting functions and sensitivities (e.g., Jacobians) exhibit an almost constant vertical shape. In contrast, the radiance sensitivities for the MHS channels tend to shift upward with higher water vapor quantity in the vertical layer where the channels are primarily sensitive. Considering that the atmospheric temperature tends to decrease with height within the troposphere, the radiance differences between the lower tropospheric AMSU-A and the MHS channels, therefore, increase with increasing peak heights of the radiance sensitivities for the MHS channels. As a result, the radiance differences between the lower tropospheric AMSU-A and the MHS channels could be used to estimate the peak height of the radiance sensitivities for the MHS channels. In the regression method, two radiance-based predictors are used: 1) the radiance difference between the AMSU-A channel 5 and the MHS channels and 2) the radiance difference between the AMSU-A channels 5 and 9. For the MHS channels 3 and 4, two predictors are employed to estimate the peak height of the water vapor Jacobian. For the MHS channel 5, two predictors are also used when the TPW exceeds 10 kg/m^2. If the TPW is less than 10 kg/m^2, however, only the radiance difference between the AMSU-A channels 5 and MHS channel 5 is employed to estimate the peak height. The TPW is empirically estimated from the AMSU-A observations.

In order to evaluate the impact of the vertical localization for the AMSU-A and MHS observations on the analysis, we conducted assimilation experiments from August 11, 2014 to September 30, 2014, using the CESM-DART DA system based on the EnKF. For the AMSU-A channels (5–11), the peak pressures, which serve as the vertical locations for the vertical localization function, are fixed. For the MHS channel 5, the peak height of the water vapor Jacobian is overestimated when TPW exceeds 10 kg/m^2, leading to the misplacement of the vertical location for the vertical localization function. For that reason, for the experiment, the vertical localization

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function is not activated for the MHS channel 5 when the TPW is larger than 10 kg/m^2. The verification results show that activating the vertical localization function for AMSU-A observations significantly enhances the analysis fields of key atmospheric variables, including temperature, zonal wind, and meridional wind. This improvement leads to a more accurate 500 hPa GPH analysis. In addition, applying the estimated peak heights of MHS channels as the vertical locations in the vertical localization function substantially improves the water vapor analysis. In particular, the water vapor analysis errors are significantly reduced in the Southern Hemisphere, where the conventional observations are relatively sparse. In addition, a substantial impact is observed in the lower troposphere by applying the vertical localization function. It means that the spurious correlation between the model variables and the observations is effectively removed by applying the vertical localization function.

Since the retrieval method proposed in this study is based on a statistically simple approach (i.e., the multivariate linear regression), it can be easily applied to other microwave instruments, provided that radiances from both temperature and water vapor channels are simultaneously available. For example, the channel characteristics of the Micro-Wave Temperature Sounder (MWTS) and the Micro-Wave Humidity Sounder (MWHS) onboard the Feng-Yun-3 (FY-3) satellite series are almost equivalent to those of the AMSU-A and MHS instruments, respectively. The proposed regression method can, therefore, be readily implemented to activate vertical localization for MWHS radiances. Additionally, this method is applicable to the Advanced Technology Microwave Sounder (ATMS) onboard the three Joint Polar Satellite System (JPSS) satellites, i.e., the Suomi National Polar-orbiting Partnership (SNPP), NOAA-20, and NOAA-21. Unlike the AMSU-A and MHS instruments, the ATMS instrument integrates both temperature and humidity sounding capabilities into a single instrument with 22 channels, consisting of temperature sounding channels (channels 1–16) and humidity sounding channels (channels 17–22). Since ATMS includes nearly all of the AMSU-A and MHS channels, the proposed method can also be applied to ATMS radiances in the EnKF-based DA system.

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